

CURRICULUM VITAE

Leshan Zhao

Office Address:

Department of Physics and Astronomy
The Johns Hopkins University
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Baltimore, MD 21218
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Education:

Johns Hopkins University	2022 – Expected 2027
Ph.D., Physics, advisor: Collin Broholm	
University of California, Irvine	2017 – 2022
B.S., Physics, advisor: Steven Barwick	

Appointments:

Advanced Sensing & Quantum Devices Intern, Nokia Bell Labs	Jun 2026 – Aug 2026
Research Assistant, Johns Hopkins University	Jun 2024 – Present
Head Teaching Assistant, Johns Hopkins University	Jun 2023 – Jun 2024
Teaching Assistant, Johns Hopkins University	Aug 2022 – Jun 2023
Research Assistant, University of California, Irvine	Feb 2020 – Aug 2022

Research Interests

- Quantum magnetism and strongly correlated electron systems
- Low-dimensional and frustrated quantum materials
- Neutron scattering
- Inverse problems and physics-informed machine learning

Awards and Grants:

Outstanding Graduating Senior Research Award — sole annual recipient	2022
Summer Undergraduate Research Grant	2020, 2021
Undergraduate Research Opportunity Grant	2020, 2021

Professional Activities:

Member: American Physical Society (Condensed Matter Physics Division, Magnetism Topical Group)

National Facilities Experience:

- Neutron Scattering
 - Oak Ridge National Laboratory (ORNL): HB-3, TOPAZ, POWGEN, SEQUOIA, CNCS
 - National Institute of Standards and Technology (NIST): MACS
- National High Magnetic Field Laboratory
 - 35 Tesla, 32 mm Bore Magnet (Cell 8)
- Brookhaven National Laboratory
 - National Synchrotron Light Source (NSLS) II: 11-ID Coherent Hard X-ray Scattering

Teaching Experience

Head Teaching Assistant, General Physics Laboratory	2023 – 2024
Teaching Assistant, General Physics Laboratory	2022 – 2023
Teaching Assistant, General Physics	2022 – 2023

Mentoring

Saebyeok Ryu, undergraduate student	2025 – Present
Govardhan Pooni, undergraduate student, now Ph.D. student at UCLA	2024 – 2026
Tingwei Liu, undergraduate student	2021 – 2022

Publications

1. L. Zhao, T. Chen, M. B. Stone, Q. Zhang, C. L. Sarkis, S. M. Koohpayeh, and C. L. Broholm, “Quenched Disorder in the Triangular Lattice Antiferromagnet $\text{YbZn}_2\text{GaO}_5$,” *Physical Review B* 113 (1), 014437 (2026). doi.org/10.1103/xn2m-1jb5
2. Z. Li, L. Zhao, C. Kakogiannis, T. P. Weihs, and J. Mueller, “3D-Printable Oil-in-Liquid-Metal Foam Emulsion,” *Matter* 8(12), 102420 (2025). doi.org/10.1016/j.matt.2025.102420
3. S. W. Barwick et al., “Refinement of Antenna Response and Ice Properties at Radio Wavelengths,” *Proceedings of the 39th International Cosmic Ray Conference, PoS(ICRC2025)984* (2025). doi.org/10.22323/1.501.0984
4. A. Anker et al. , “Developing New Analysis Tools for Near Surface Radio-Based Neutrino Detectors,” *Journal of Cosmology and Astroparticle Physics* 2023(10), 060 (2023). 10.1088/1475-7516/2023/10/060
5. J. A. Aguilar et al. , “Triboelectric Backgrounds to Radio-Based Polar Ultra-High Energy Neutrino Experiments,” *Astroparticle Physics* 145, 102790 (2023). doi.org/10.1016/j.astropartphys.2022.102790

6. S.-H. Wang et al., “TAROGEM: Radio Antenna Array on Antarctic High Mountain for Detecting Near-Horizontal Ultra-High Energy Air Showers,” *Journal of Cosmology and Astroparticle Physics* 2022(11), 022 (2022), doi.org/10.1088/1475-7516/2022/11/022
7. A. Anker et al., “Measuring the Polarization Reconstruction Resolution of the ARIANNA Neutrino Detector with Cosmic Rays,” *Journal of Cosmology and Astroparticle Physics* 2022(04), 022 (2022). doi.org/10.1088/1475-7516/2022/04/022
8. A. Anker et al., “Improving Sensitivity of the ARIANNA Detector by Rejecting Thermal Noise with Deep Learning,” *Journal of Instrumentation* 17(03), P03007 (2022). doi.org/10.1088/1748-0221/17/03/P03007
9. J. Beise et al., “Development of an In-Situ Calibration Device of Firn Properties for Askaryan Neutrino Detectors,” *Proceedings of the 37th International Cosmic Ray Conference, PoS(ICRC2021)1069* (2022). doi.org/10.22323/1.395.1069
10. L. Zhao et al., “Polarization Reconstruction of Cosmic Rays with the ARIANNA Neutrino Radio Detector,” *Proceedings of the 37th International Cosmic Ray Conference, PoS(ICRC2021)1156* (2022). doi.org/10.22323/1.395.1156
11. A. Anker et al., “A Novel Trigger Based on Neural Networks for Radio Neutrino Detectors,” *Proceedings of the 37th International Cosmic Ray Conference, PoS(ICRC2021)1074* (2022). doi.org/10.22323/1.395.1074
12. S. W. Barwick et al., “Capabilities of ARIANNA: Neutrino Pointing Resolution and Implications for Future Ultra-High Energy Neutrino Astronomy,” *Proceedings of the 37th International Cosmic Ray Conference, PoS(ICRC2021)1151* (2022). doi.org/10.22323/1.395.1151
13. S. W. Barwick et al., “Science Case and Detector Concept for the ARIANNA High Energy Neutrino Telescope at Moore’s Bay, Antarctica,” *Proceedings of the 37th International Cosmic Ray Conference, PoS(ICRC2021)1190* (2022). doi.org/10.22323/1.395.1190
14. S.-H. Wang et al., “TAROGEM: Radio Observatory on Antarctic High Mountain for Detecting Near-Horizon Ultra-High Energy Air Showers,” *Proceedings of the 37th International Cosmic Ray Conference, PoS(ICRC2021)1173* (2022). doi.org/10.22323/1.395.1173

Talks & Presentations

1. “Quasiparticle Interactions in the Spin-1/2 Heisenberg Spin Chain CuPzN.” American Physical Society Global Physics Summit, Denver, Colorado, 2026. (Contributed talk)
2. “Quenched Disorder in Quantum Spin Liquid Candidate YbZn₂GaO₅.” Center for High Resolution Neutron Scattering Summer School on Neutron Spectroscopy, National Institute of Standards and Technology, Gaithersburg, Maryland, 2025. (Contributed talk)
3. “Novel Background Rejection Techniques in a Search for Ultra-High Energy Neutrinos Using an ARIANNA Detector Station at the South Pole.” Neutrino 2022, Seoul, South Korea, July 2022. (Poster)
4. “Measuring the Polarization Reconstruction Resolution of the ARIANNA Neutrino Detector with Cosmic Rays.” Neutrino 2022, Seoul, South Korea, July 2022. (Poster)
5. “Polarization Reconstruction of Cosmic Rays with the ARIANNA Neutrino Radio Detector.” 37th International Cosmic Ray Conference, Berlin, Germany, July 2021. (Contributed talk)